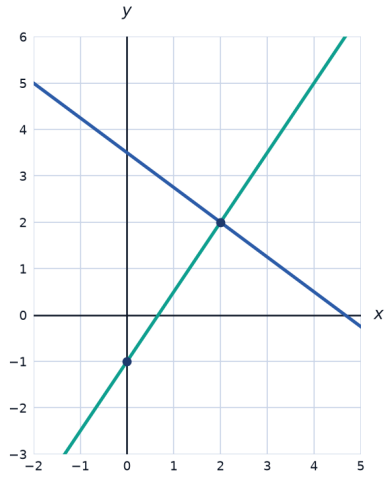
Matrices & Vectors

Section 1: Matrix Multiplication & Inverses

1 If A is a 3×2 matrix and B is a 2×4 matrix, what are the dimensions of the product matrix AB?



a. 3×4

b. 2×2

c. 4×3

d. Multiplication is undefined

2 What is the inverse of the matrix $M = [[2, 5], [1, 3]]$?

a. $[[3, -5], [-1, 2]]$

b. $[[-3, 5], [1, -2]]$

c. $[[2, -5], [-1, 3]]$

d. $[[3, 1], [5, 2]]$

3 Let A and B be invertible $n \times n$ matrices. Which expression represents $(AB)^{-1}$?

a. $B^{-1}A^{-1}$

b. $A^{-1}B^{-1}$

c. AB^{-1}

d. BA^{-1}

Section 2: Determinants & Cramer's Rule

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4 What is the determinant of the 3×3 matrix $[[1, 2, 3], [0, 4, 5], [0, 0, 6]]$?

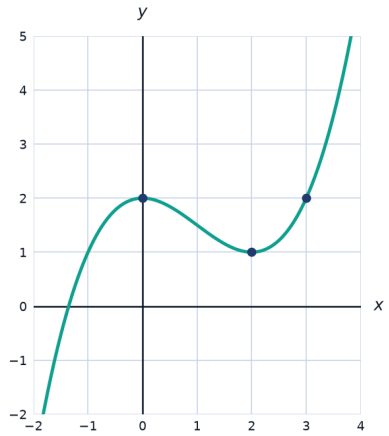
a. 24

b. 30

c. 36

d. 0

5 If the determinant of a 2×2 matrix A is 5, what is $\det(3A)$?



a. 15

b. 45

c. 75

d. 135

6 For what value of x is the matrix $[[x, 3], [12, x]]$ singular?

a. ± 6

b. ± 4

c. ± 12

d. 0

Section 3: Dot Product & Angle Between Vectors

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7 What is the angle θ between vectors $\mathbf{u} = \langle 1, 0 \rangle$ and $\mathbf{v} = \langle 1, 1 \rangle$?

a. 30°

b. 45°

c. 60°

d. 90°

8 For what value of k are the vectors $\mathbf{a} = \langle 4, k \rangle$ and $\mathbf{b} = \langle 3, -6 \rangle$ orthogonal?

a. 1

b. 2

c. 3

d. 4

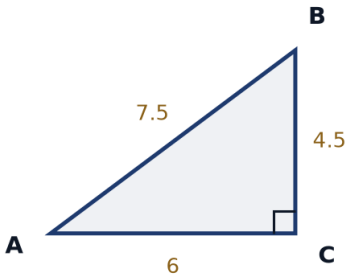
9 What is the scalar projection of vector $\mathbf{u} = \langle 4, 3 \rangle$ onto vector $\mathbf{v} = \langle 12, 5 \rangle$?

a. $\frac{63}{13}$

b. 5

c. $\frac{48}{13}$

d. $\frac{15}{13}$



Section 4: 3D Vectors & Cross Product

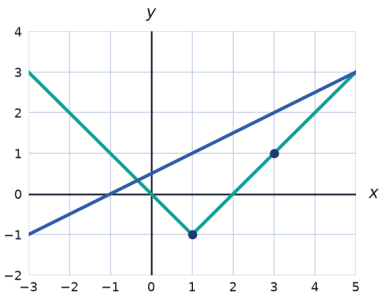
10 What is the cross product $\mathbf{i} \times \mathbf{j}$ in three-dimensional space?

a. $\mathbf{k}$

b. $-\mathbf{k}$

c. $\mathbf{0}$

d. 1



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11 What is the magnitude of the vector $\mathbf{v} = \langle 2, -3, 6 \rangle$?

a. 7

b. 9

c. $\sqrt{49}$

d. 8

12 What is the area of the parallelogram determined by vectors $\mathbf{a} = \langle 3, 0, 0 \rangle$ and $\mathbf{b} = \langle 0, 4, 0 \rangle$?

a. 6

b. 12

c. 24

d. 7

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